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Supplement to "Variable Length Memory Chains: characterization of stationary probability measures"

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Abstract This supplementary material contains most of the proofs of the theorems and lemma appearing in Cénac et al. (2020), together with an appendix giving an example of invariant σ-finite measure that charges irrational infinite contexts.

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1. Proofs

1.1. Proofs of Section 2 (general case)

Proof of lemma 2.15 (Cascade formulae).

(i) Assume first that $\pi(w\mathcal{R}) \neq 0$. Then, since w is noninternal, $\mathrm{cont}(w)$ is well defined so that, by stationarity,

$$\begin{aligned}\pi(\alpha w\mathcal{R}) &= \mathbf{P}_\pi(U_1 \in \alpha w\mathcal{R}) \\ &= \mathbf{P}_\pi(U_1 \in \alpha w\mathcal{R} | U_0 \in w\mathcal{R})\,\mathbf{P}_\pi(U_0 \in w\mathcal{R}) \\ &= q_{\mathrm{cont}(w)}(\alpha)\pi(w\mathcal{R})\end{aligned}$$

proving (2.15). If $\pi(w\mathcal{R}) = 0$, then, by stationarity,

$$\pi(\alpha w\mathcal{R}) = \mathbf{P}_\pi(U_1 \in \alpha w\mathcal{R}) \leqslant \mathbf{P}_\pi(U_0 \in w\mathcal{R}) = 0$$

so that (2.15) remains true.

(ii) By an argument similar to the one above, if $\pi(r) = 0$, then

$$\pi(\alpha r) = \mathbf{P}_\pi(U_1 = \alpha r) \leqslant \mathbf{P}_\pi(U_0 = r) = \pi(r) = 0.$$

If, on the contrary, $\pi(r) \neq 0$, then

$$\pi(\alpha r) = \mathbf{P}_\pi(U_1 = \alpha r) = \mathbf{P}_\pi(U_1 = \alpha r | U_0 = r)\,\mathbf{P}_\pi(U_0 = r) = q_{\mathrm{cont}(r)}(\alpha)\pi(r),$$

which proves (ii).

(iii) Direct induction from Formula (2.15). □

Proof of lemma 2.16. (i) We prove that if w is a finite word and if $\alpha \in \mathcal{A}$, then $[\pi(\alpha w\mathcal{R}) = 0] \Rightarrow [\pi(w\mathcal{R}) = 0]$. An induction on the length of w is then sufficient to prove the result since $\pi(\mathcal{R}) = 1$. Remember that $\mathcal{I}$ stands for the set of finite internal words.

1. Assume that $w \notin \mathcal{I}$ and that $\pi(\alpha w\mathcal{R}) = 0$. Then, as a consequence of the cascade formula (2.15), $0 = \pi(w\mathcal{R})\, q_{\mathrm{cont}(w)}(\alpha)$. As no q_c vanishes, $\pi(w\mathcal{R}) = 0$.
2. Assume now $w \in \mathcal{I}$ and $\pi(\alpha w\mathcal{R}) = 0$. Then, by disjoint union and Lemma 2.15, since π is stationary,

$$0 = \pi(\alpha w\mathcal{R}) = \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots}} \pi(\alpha c\mathcal{R}) + \sum_{\substack{c\in\mathcal{C}^i \\ c=w\cdots}} \pi(\alpha c) = \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots}} \pi(c\mathcal{R})\, q_c(\alpha) + \sum_{\substack{c\in\mathcal{C}^i \\ c=w\cdots}} \pi(c)\, q_c(\alpha).$$

As no q_c vanishes, all the $\pi(c\mathcal{R})$ and the $\pi(c)$ necessarily vanish so that, by disjoint union,

$$\pi(w\mathcal{R}) = \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots}} \pi(c\mathcal{R}) + \sum_{\substack{c\in\mathcal{C}^i \\ c=w\cdots}} \pi(c) = 0.$$

(ii) Denote $r = \alpha_1\alpha_2\cdots$ and $r_n = \alpha_n\alpha_{n+1}\cdots$ its n-th suffix, for every $n \geqslant 1$. Since π is stationary, an elementary induction from Formula (8) implies that, for every $m \geqslant 1$,

$$\pi(r) = \left(\prod_{k=1}^{m} q_{\mathrm{cont}(r_{k+1})}(\alpha_k)\right)\pi(r_{m+1}). \tag{1}$$

1. Assume first that $r = st^\infty$ is ultimately periodic, where s and t are finite words, $t = \beta_1 \cdots \beta_T$ being nonempty. Then, because of (1), $\pi(r) \leqslant \pi(t^\infty)$ and $\pi(t^\infty) = \rho\pi(t^\infty)$ where
$$\rho = \prod_{k=1}^{T} q_{\mathrm{cont}(\beta_{k+1}\cdots\beta_T t^\infty)}(\beta_k).$$
In this product, the term obtained for $k = T$ writes $q_{\mathrm{cont}(t^\infty)}(\beta_T)$. Since the probability measures q_c are all assumed never to vanish, they cannot take 1 as a value so that $\rho < 1$, which implies that $\pi(t^\infty) = 0$. Note that this argument proves that an invariant probability measure π vanishes on ultimately periodic infinite words as soon as the q_c never take 1 as a value (this assumption is weaker than non-nullness).
2. Assume on the contrary that r is aperiodic. Then, $m \neq n \Longrightarrow r_n \neq r_m$ for all $n, m \geqslant 1$: the r_n are all distinct among the infinite branches of the context tree. Thus, by disjoint union,
$$\sum_{n\geqslant 1} \pi(r_n) \leqslant \sum_{c\in\mathcal{C}^i} \pi(c) \leqslant \pi(\mathcal{R}) = 1,$$
which implies in particular that $\pi(r_n)$ tends to 0 when n tends to infinity. Since $\pi(r) \leqslant \pi(r_n)$ because of Formula (1), this leads directly to the result.

□

Proof of theorem 2.18. Proof of *(i)*. If π is a stationary probability measure, disjoint union, Lemma 2.16(ii) and the Cascade Formula (9) imply that
$$1 = \sum_{c\in\mathcal{C}^f} \pi(c\mathcal{R}) = \sum_{c\in\mathcal{C}^f} \mathrm{casc}(c)\pi(\alpha_c s_c \mathcal{R}).$$
Gathering together all the contexts that have the same alpha-LIS leads to
$$1 = \sum_{\alpha s\in\mathcal{S}} \pi(\alpha s\mathcal{R}) \left(\sum_{c\in\mathcal{C}^f,\ c=\cdots[\alpha s]} \mathrm{casc}(c) \right).$$
Now, by Lemma 2.16(i), $\pi(\alpha s\mathcal{R}) \neq 0$ for all $\alpha s \in \mathcal{S}$. This forces the sums of cascades to be finite.

Proof of *(ii)*.

1) *Injectivity.* Let π be a stationary probability measure on $\mathcal{R}$. As the cylinders based on finite words generate the whole σ-algebra, π is determined by the $\pi(w\mathcal{R})$, $w \in \mathcal{W}$. Now write any $w \in \mathcal{W} \setminus \{\emptyset\}$ as $w = p\alpha s$ where αs is the alpha-LIS of w and $p \in \mathcal{W}$ (beware, αs may not be the alpha-LIS of a context). As π is stationary, the cascade formula (9) entails $\pi(w\mathcal{R}) = \mathrm{casc}(w)\pi(\alpha s\mathcal{R})$. As a consequence, π is determined by its values on the words αs where $s \in \mathcal{I}$ is internal and $\alpha \in \mathcal{A}$. Now, as $s \in \mathcal{I}$, by disjoint union, cascade formula (2.15) and Lemma 2.16(ii),
$$\pi(\alpha s\mathcal{R}) = \sum_{c\in\mathcal{C}^f,\ c=s\cdots} \pi(c\mathcal{R})\, q_c(\alpha).$$

This means that π is in fact determined by the $\pi(c\mathcal{R})$ where c is a finite context. Lastly, as above, the stationarity of π, the cascade formula (9) and the decomposition of any context c into $c = p_c\alpha_c s_c$ where $\alpha_c s_c$ is the alpha-LIS of c together imply that π is determined by the $\pi(\alpha s\mathcal{R})$ where $s \in \mathcal{S}$ (remember, $\mathcal{S}$ denotes the set of all alpha-LIS *of contexts*). This proves that the restriction of f to stationary measures is one-to-one.

2) *Image of a stationary probability measure.* Let $\pi \in \mathcal{M}_1(\mathcal{R})$ be stationary. By disjoint union, as above, if $\alpha s \in \mathcal{S}$,

$$\pi(\alpha s\mathcal{R}) = \sum_{c\in\mathcal{C}^f,\ c=s\cdots} \pi(c\mathcal{R})\, q_c(\alpha).$$

Applying the Cascade Formula (9) to all contexts in the sum and noting that $\mathrm{casc}(\alpha c) = q_c(\alpha)\,\mathrm{casc}(c)$, one gets

$$\pi(\alpha s\mathcal{R}) = \sum_{c\in\mathcal{C}^f,\ c=s\cdots} \mathrm{casc}(\alpha c)\pi(\alpha_c s_c\mathcal{R}).$$

Gathering together all the contexts that have the same alpha-LIS entails

$$\pi(\alpha s\mathcal{R}) = \sum_{\beta t\in\mathcal{S}} \pi(\beta t\mathcal{R}) \left(\sum_{\substack{c\in\mathcal{C}^f \\ c=s\cdots=\cdots[\beta t]}} \mathrm{casc}(\alpha c) \right) = \sum_{\beta t\in\mathcal{S}} \pi(\beta t\mathcal{R})\, Q_{\beta t,\alpha s}.$$

This means that the row vector $(\pi(\alpha s\mathcal{R}))_{\alpha s\in\mathcal{S}}$ is a left-fixed vector for the matrix Q. We have shown that f sends a stationary probability measure to a left-fixed vector for Q with positive entries. Moreover, as in the proof of (i), Equality (2.18) holds.

3) *Surjectivity.*

Let $(v_{\alpha s})_{\alpha s\in\mathcal{S}} \in [0,+\infty[^{\mathcal{S}}$ be a row vector, left-fixed by Q, that satisfies $\sum_{\alpha s\in\mathcal{S}} v_{\alpha s}\kappa_{\alpha s} = 1$. Let μ be the function defined on $\mathcal{S}$ by $\mu(\alpha s) = v_{\alpha s}$. Denoting by $\alpha_w s_w$ the alpha-LIS of any finite non-empty word w, the function μ extends to any finite non-empty word in the following way:

$$\forall w \in \mathcal{W}\setminus\{\emptyset\},\ \mu(w) = \mathrm{casc}(w) \sum_{\substack{c\in\mathcal{C}^f \\ c=s_w\cdots}} \mathrm{casc}(\alpha_w c)\mu(\alpha_c s_c) \in [0,+\infty]. \tag{2}$$

Notice that this definition actually extends μ because of the fixed vector property, and that, at this moment of the proof, $\mu(w)$ could be infinite. Notice also that this implies $\mu(w) = \mathrm{casc}(w)\mu(\alpha_w s_w)$ for any $w \in \mathcal{W}$, $w \neq \emptyset$.

For every $n \geqslant 1$ and for all $w \in \mathcal{W}$ such that $|w| = n$, define $\pi_n(w) = \mu(w)$. This clearly defines a $[0,+\infty]$-valued measure π_n on $\mathcal{A}_n = \prod_{1\leqslant k\leqslant n}\mathcal{A}$. Besides, π_1 is a probability measure. Indeed, because of Definition (2) and Remark (2.11),

$$\sum_{a\in\mathcal{A}} \mu(a) = \sum_{c\in\mathcal{C}^f} \left(\sum_{a\in\mathcal{A}} \mathrm{casc}(ac) \right) \mu(\alpha_c s_c) = \sum_{c\in\mathcal{C}^f} \mathrm{casc}(c)\,\mu(\alpha_c s_c)$$

which can be written

$$\sum_{a\in\mathcal{A}}\mu(a)=\sum_{\alpha s\in\mathcal{S}}\mu\left(\alpha s\right)\sum_{\substack{c\in\mathcal{C}^f\\ c=\cdots[\alpha s]}}\mathrm{casc}(c)=\sum_{\alpha s\in\mathcal{S}}v_{\alpha s}\kappa_{\alpha s}=1,$$

the last equality coming from the assumption on $(v_{\alpha s})_{\alpha s}$.

In view of applying Kolmogorov extension theorem, the consistency condition states as follows: $\pi_{n+1}(w\mathcal{A})=\pi_n(w)$ for any $w\in\mathcal{W}$ of length n. This is true because

$$\sum_{a\in\mathcal{A}}\mu(wa)=\mu(w). \tag{3}$$

Indeed, for any $a\in\mathcal{A}$, since s_w is internal, s_wa is either internal or a context. Furthermore,

- if $s_wa\in\mathcal{I}$ then $s_{wa}=s_wa$, $\alpha_{wa}=\alpha_w$ and $\mathrm{casc}(wa)=\mathrm{casc}(w)$ so that
$$\mu(wa)=\mathrm{casc}(w)\sum_{\substack{c\in\mathcal{C}^f\\ c=s_wa\cdots}}\mathrm{casc}\left(\alpha_wc\right)\mu\left(\alpha_cs_c\right); \tag{4}$$
- if $s_wa\in\mathcal{C}$ then denote $\kappa=s_wa$ so that $\mathrm{casc}(wa)=\mathrm{casc}(w)\,\mathrm{casc}\left(\alpha_w\kappa\right)$, $\alpha_{wa}=\alpha_\kappa$ and $s_{wa}=s_\kappa$. Thus,
$$\mu(wa)=\mathrm{casc}(wa)\sum_{c=s_\kappa\cdots}\mathrm{casc}\left(\alpha_\kappa c\right)\mu\left(\alpha_cs_c\right)=\mathrm{casc}(w)\,\mathrm{casc}\left(\alpha_w\kappa\right)\mu\left(\alpha_\kappa s_\kappa\right),$$
which implies that (4) still holds, the sum being reduced to one single term since s_wa is itself a context.

Valid in all cases, Formula (4) easily implies Claim (3). Consequently all the π_n are probability measures. By Kolmogorov extension theorem, there exists a unique probability measure π on $\mathcal{R}$ such that $\pi_{|\mathcal{A}_n}=\pi_n$ for every n. Note that this result implies that $v_{\alpha s}=\pi\left(\alpha s\mathcal{R}\right)\leqslant 1$, for every $\alpha s\in\mathcal{S}$.

Furthermore, $\pi(c)=0$ for any infinite context c. Indeed, one has successively,

$$\begin{aligned}1&=\sum_{c\in\mathcal{C}^f}\pi\left(c\mathcal{R}\right)+\sum_{c\in\mathcal{C}^i}\pi\left(c\right)\\&=\sum_{c\in\mathcal{C}^f}\mu(c)+\sum_{c\in\mathcal{C}^i}\pi\left(c\right)\end{aligned}$$

and, besides,

$$\sum_{c\in\mathcal{C}^f}\mu(c)=\sum_{c\in\mathcal{C}^f}\mathrm{casc}(c)\mu\left(\alpha_cs_c\right)=\sum_{\alpha s\in\mathcal{S}}\mu\left(\alpha s\right)\sum_{\substack{c\in\mathcal{C}^f\\ c=\cdots[\alpha s]}}\mathrm{casc}(c)=\sum_{\alpha s\in\mathcal{S}}v_{\alpha s}\kappa_{\alpha s}=1$$

so that $\sum_{c\in\mathcal{C}^i}\pi\left(c\right)=0$.

Finally, the stationarity of π follows from the identity $\sum_{a\in\mathcal{A}}\mu(aw)=\mu(w)$ for any finite word w. Namely:

- if $w \notin \mathcal{I}$, then for $a \in \mathcal{A}$, $s_{aw} = s_w$, $\alpha_{aw} = \alpha_w$ hence

$$\sum_{a\in\mathcal{A}} \mu(aw) = \left(\sum_{a\in\mathcal{A}} \mathrm{casc}(aw)\right) \sum_{\substack{c\in\mathcal{C}^f \\ c=s_w\cdots}} \mathrm{casc}\left(\alpha_w c\right) \mu\left(\alpha_c s_c\right).$$

  Now, Remark 2.11 entails the claim.
- if $w \in \mathcal{I}$, then for $a \in \mathcal{A}$, $s_{aw} = w$, $\alpha_{aw} = a$ and $\mathrm{casc}(aw) = 1$ thus

$$\mu(aw) = \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots}} \mathrm{casc}(ac)\mu\left(\alpha_c s_c\right).$$

  Using again Remark 2.11, it comes

$$\begin{aligned}
\sum_{a\in\mathcal{A}} \mu(aw) &= \sum_{c\in\mathcal{C}^f,\ c=w\cdots} \left(\sum_{a\in\mathcal{A}} \mathrm{casc}(ac)\right) \mu(\alpha_c s_c) \\
&= \sum_{c\in\mathcal{C}^f,\ c=w\cdots} \mathrm{casc}(c)\mu(\alpha_c s_c) \\
&= \sum_{c\in\mathcal{C}^f,\ c=w\cdots} \mu(c) \\
&= \sum_{c\in\mathcal{C}^f,\ c=w\cdots} \pi\left(c\mathcal{R}\right) \\
&= \pi\left(w\mathcal{R}\right) - \sum_{\substack{c\in\mathcal{C}^i \\ c=w\cdots}} \pi\left(c\right) = \mu(w),
\end{aligned}$$

the last equality being valid because π vanishes on infinite contexts. Since $f(\pi) = (v_{\alpha s})_{\alpha s}$, this concludes the proof. □

1.2. Proofs of Section 3 (stable case)

Proof of Proposition 3.1. $(i) \implies (ii)$. Take $c \in \mathcal{C}$ and $\alpha \in \mathcal{A}$. Assume that $\alpha c \in \mathcal{I}$. So, if $\beta \in \mathcal{A}$ is any letter, then $\alpha c\beta \in \mathcal{T}$. The item (i) implies therefore that $c\beta \in \mathcal{T}$, which contradicts $c \in \mathcal{C}$.

$(ii) \implies (i)$. Take $\alpha \in \mathcal{A}$ and $w \in \mathcal{W}$ such that $\alpha w \in \mathcal{T}$. If $w \notin \mathcal{T}$ then there exists a finite context c such that $w = cw'$ with $w' \neq \emptyset$. It comes $\alpha cw' \in \mathcal{T}$, which implies $\alpha c \in \mathcal{I}$ and this contradicts (ii).

$(i) \iff (iii)$ is straightforward.

$(ii) \implies (iv)$ What needs to be proved is that $\mathrm{cont}\left(U_{n+1}\right)$ only depends on U_n through $\mathrm{cont}\left(U_n\right)$. In other words, we shall prove that for all $s \in \mathcal{R}, \alpha \in \mathcal{A}$, $\mathrm{cont}\left(\alpha s\right)$ only depends on s through $\mathrm{cont}(s)$. This is clear because

$$\mathrm{cont}(\alpha s) = \mathrm{cont}\left(\alpha\, \mathrm{cont}(s)\right).$$

Indeed, $\mathrm{cont}(s) \in \mathcal{C}$ and if $\mathrm{cont}(s) \in \mathcal{C}^f$, then (ii) implies $\alpha\,\mathrm{cont}(s) \notin \mathcal{I}$ (if $\mathrm{cont}(s)$ is an infinite context, this means that $s = \mathrm{cont}(s)$ and the above equality is straightforward). Therefore $\alpha\,\mathrm{cont}(s)$ writes cw with $c \in \mathcal{C}$ and $w \in \mathcal{W}$. On one hand, this entails $\mathrm{cont}\,(\alpha\,\mathrm{cont}(s)) = c$. On the other hand, this means that cw is a prefix of αs thus $\mathrm{cont}(\alpha s) = c$.

$(iv) \implies (ii)$ We shall prove the contrapositive. Assume there exists $c \in \mathcal{C}^f$ and $\alpha \in \mathcal{A}$ such that $\alpha c \in \mathcal{I}$. Let $s \in \mathcal{R}$ such that $\mathrm{cont}(s) = c$. As $\alpha c \in \mathcal{I}$, $\mathrm{cont}(\alpha s)$ is a context which has αc as a strict prefix. Therefore, $\mathrm{cont}(\alpha s)$ does not only depend on $\mathrm{cont}(s)$, but on a prefix of s strictly longer than $\mathrm{cont}(s)$. Thus, conditionally to C_n, the transition to the context C_{n+1} does not depend on C_n but on a strictly longer prefix of U_n. This proves that $(C_n)_n$ is not Markovian. □

Proof of Lemma 3.4. 1. Assume that t is a context LIS such that $c = t\cdots$, then, by definition of a context LIS, there exists $\alpha \in \mathcal{A}$ such that αt is a context alpha-LIS. Since $\mathcal{T}$ is stable, Lemma 3.3 implies that αt is a context, therefore $\alpha c \notin \mathcal{C}$ because two different contexts cannot be prefix of one another. Thus $\mathcal{A}_c \neq \emptyset$.

2. Let α be in $\mathcal{A}_c$ so that αc is not a context. As $\mathcal{T}$ is stable, αc is not internal, thus it is an external node. Let $c' = \mathrm{cont}(\alpha c)$. The context c' is a strict prefix of αc. Since $\mathcal{T}$ is stable, $\sigma(c')$ is non-external. But $\sigma(c')$ cannot be a context because it is a prefix of c which is a context. Thus $\sigma(c')$ is internal. This implies that $t_\alpha := \sigma(c')$ is the LIS of c' and a prefix of c as well, and that $\alpha t_\alpha = c' \in \mathcal{C}$. Besides, whenever (i) and (ii) are satisfied, t_α writes necessarily $t_\alpha = \sigma\,(\mathrm{cont}\,(\alpha c))$. Thus existence and uniqueness of t_α are proven. Finally, since $c' = \alpha t_\alpha = \mathrm{cont}(\alpha c)$, t_α is a prefix of c, so that for every $\beta \notin \mathcal{A}_c$, βt_α is a prefix of the context βc. Consequently, $\beta t_\alpha \notin \mathcal{C}$ because two different contexts cannot be prefix of one another. □

Proof of Lemma 3.6. *Irreducibility.* Let c and c' be finite contexts. Denote $c' = \alpha_k \dots \alpha_2 \alpha_1$, $k \geqslant 1$. In order to prove that the Markov chain (C_n) has a non-null transition from c to c', let us add the successive letters of c' (starting from α_1) to the left of c and prove that at each time, the transition is possible and non-null. Assume that $C_0 = c$ and consider the word $\alpha_1 c$. As $\mathcal{T}$ is shift-stable $\alpha_1 c \notin \mathcal{I}$ and $c_1 := \mathrm{cont}(\alpha_1 c)$ is a (possibly not strict) prefix of $\alpha_1 c$ which may be written $c_1 = \alpha_1 w_1$ with w_1 a prefix (possibly empty) of c. The transition equals $q_c(\alpha_1)$ and is therefore non-null. Let us add the second letter and consider the word $\alpha_2\alpha_1 w_1 = \alpha_2 c_1$. Again $\alpha_2\alpha_1 w_1 \notin \mathcal{I}$ and $c_2 := \mathrm{cont}\,(\alpha_2\alpha_1 w_1)$ is a prefix of $\alpha_2\alpha_1 w_1$. Here, the point is that c_2 cannot be α_2 otherwise we would have $\sigma^{k-2}(c') = \alpha_2\alpha_1 \notin \mathcal{T}$ which would contradict the shift-stability of $\mathcal{T}$. Therefore $c_2 = \alpha_2\alpha_1 w_2$ with w_2 a prefix of w_1. By adding successively the letters of c', with the same arguments, one gets a sequence of contexts $c_j = \mathrm{cont}\,(\alpha_j c_{j-1}) = \alpha_j\alpha_{j-1}\dots\alpha_1\dots$. The last step necessarily writes $c_k = \alpha_k\dots\alpha_1 = c'$ because c' is a context. At each step, the transition is non zero because all the $q_c(\alpha)$ are non-null.

Aperiodicity. Let us prove that, given $c \in \mathcal{C}^f$, the g.c.d. of the lengths of the admissible paths from c to itself equals 1. Let k be the length of c. The above proof of the irreducibility shows that there exists an admissible path of length k joining c to itself. Using similar arguments, let α be any letter and let c' be the context $c' = \mathrm{cont}\,(\alpha c)$. There is

an admissible path of length 1 joining c to c'. Using again the proof of the irreducibility, there is also an admissible path of length k joining c' to c. Combining these two paths provides a path of length $k+1$ that joins c to itself. Since $gcd(k, k+1) = 1$, the chain is aperiodic. □

Proof of Proposition 3.13. The initial context $C_0 = \mathrm{cont}\,(U_0)$ is assumed to be finite. Since the context tree is stable, this implies that all $C_n = \mathrm{cont}\,(U_n)$ are also almost surely finite words – see Proposition 3.1(ii). Thanks to this fact, the definition of $(Z_n)_n$ makes sense (see (12)).

Let us prove (i), i.e. that T_n is almost surely finite (and S_n as well), by induction on $n \geqslant 1$. Remember that $S_0 = 0$. To lighten the computation, assume that C_0 is a context alpha-LIS. If not, C_0 writes $\beta_1 \dots \beta_p \alpha s$ and a term $\mathrm{casc}(C_0)$ has to be added to the successive equalities without modifying the argumentation.

$$\mathbf{P}(T_1 = +\infty) = \sum_{\alpha s \in \mathcal{S}} \mathbf{P}(T_1 = +\infty | C_0 = \alpha s)\, \mathbf{P}(C_0 = \alpha s).$$

It is sufficient to prove that, for all $\alpha s \in \mathcal{S}$, $\mathbf{P}(T_1 = +\infty | C_0 = \alpha s) = 0$. Now

$$\mathbf{P}(T_1 = +\infty | C_0 = \alpha s) = \lim_{k \to \infty} \mathbf{P}(T_1 \geqslant k | C_0 = \alpha s).$$

With the description of the process in Section 3.2.2, see also Figure 2,

$$\begin{aligned}
\mathbf{P}\left(T_1 \geqslant k | C_{S_0} = \alpha s\right) &= \sum_{\substack{\beta_1, \dots, \beta_{k-1} \in \mathcal{A} \\ \forall i \leqslant k-1,\ \beta_i \dots \beta_1 \alpha s \in \mathcal{C}}} q_{\alpha s}(\beta_1)\, q_{\beta_1 \alpha s}(\beta_2) \dots q_{\beta_{k-2} \dots \beta_1 \alpha s}(\beta_{k-1}) \\
&= \sum_{\substack{\beta_1, \dots, \beta_{k-1} \in \mathcal{A} \\ \forall i \leqslant k-1,\ \beta_i \dots \beta_1 \alpha s \in \mathcal{C}}} \mathrm{casc}\,(\beta_{k-1} \dots \beta_1 \alpha s) \\
&= \sum_{\substack{c \in \mathcal{C},\ c = \cdots [\alpha s] \\ |c| = |\alpha s| + k - 1}} \mathrm{casc}\, c \\
&= \kappa_{\alpha s}(k),
\end{aligned}$$

which, by assumption, tends to 0 when k tends to infinity. Consequently, T_1 and $S_1 = T_1$ is a.s. finite.

Now, assume that, for all $n \geqslant 1$, S_{n-1} is a.s. finite. Repeat the above argument, replacing S_0 by S_{n-1} and T_1 by T_n. It appears that for all $n \geqslant 1$,

$$\mathbf{P}\left(T_1 \geqslant k | C_{S_0} = \alpha s\right) = \mathbf{P}\left(T_n \geqslant k | C_{S_{n-1}} = \alpha s\right) = \kappa_{\alpha s}(k),$$

so that T_n is a.s. finite and S_n as well. Note that this proves in passing that the T_n are almost surely finite if and only if all the $\kappa_{\alpha s}(k)$ tend to 0 when k tends to infinity, which has been evoked in the description of the process $(Z_n)_n$, a few lines before Proposition 3.13's statement.

Remembering the description of the process at the beginning of Section 3.2.2, (ii) is straightforward. Moreover, $C_{S_{n-1}} = J_{n-1}$ so that summing on k gives $\mathbf{E}\left(T_n \middle| J_{n-1} = \alpha s\right) = \kappa_{\alpha s}$. This makes the proof of (i) complete.

For (iii), based on the description of the process and the finiteness of the T_n and S_n, it is clear that the distribution of (J_{n+1}, T_{n+1}) conditioned on the past only depends on (J_n, T_n), so that $(J_n, T_n)_{n\geqslant 0}$ is a Markov process. For $j \geqslant 0$, $k \geqslant 2$, $\alpha s \in \mathcal{S}$, $\beta t \in \mathcal{S}$,

$$\mathbf{P}\left(J_{n+1} = \beta t, T_{n+1} = k \middle| J_n = \alpha s, T_n = j\right) = \mathbf{P}\left(Z_{S_n+k} = \beta t, Z_{S_n+k-1} = \cdots = Z_{S_n+1} = \alpha s \middle| Z_{S_n} = \alpha s\right).$$

Notice that this expression does not depend on j, ensuring that $(J_n, T_n)_{n\geqslant 0}$ is a Markov renewal chain. Continuing the computation leads to

$$\begin{aligned}
&\mathbf{P}\left(J_{n+1} = \beta t, T_{n+1} = k \middle| J_n = \alpha s, T_n = j\right) = \mathbf{P}\left(J_{n+1} = \beta t, T_{n+1} = k \middle| J_n = \alpha s\right) \\
&= \sum_{\beta_1,\ldots,\beta_{k-1}} \mathbf{P}\left(U_{S_n+k} = \beta t \cdots, U_{S_n+k-1} = \beta_{k-1}\cdots\beta_1\alpha s \cdots, \ldots, U_{S_n+1} = \beta_1\alpha s \cdots \middle| U_{S_n} = \alpha s \cdots\right) \\
&= \sum_{\beta_1,\ldots,\beta_{k-1}} q_{\beta_{k-1}\cdots\beta_1\alpha s}(\beta)\ldots q_{\beta_1\alpha s}(\beta_2) q_{\alpha s}(\beta_1),
\end{aligned}$$

where the sum concerns the letters $\beta_1, \ldots, \beta_{k-1}$ such that $c = \beta_{k-1}\cdots\beta_1\alpha s$ is a context that begins with the LIS t, and βc is not a context. This can be shortly written under the form

$$\mathbf{P}\left(J_{n+1} = \beta t, T_{n+1} = k \middle| J_n = \alpha s, T_n = j\right) = \sum_{\substack{c\in\mathcal{C},\ c=t\ldots \\ c=\ldots[\alpha s] \\ |c|=|\alpha s|+k-1}} \operatorname{casc}(\beta c). \tag{5}$$

For $k = 1$, the calculation reduces to the following: let $j \geqslant 0$, $\alpha s \in \mathcal{S}$, $\beta t \in \mathcal{S}$ and assume that αs begins with the LIS t. Then

$$\mathbf{P}\left(J_{n+1} = \beta t, T_{n+1} = 1 \middle| J_n = \alpha s, T_n = j\right) = \mathbf{P}\left(Z_{S_n+1} = \beta t \middle| Z_{S_n} = \alpha s\right) = q_{\alpha s}(\beta)$$

which equals $\operatorname{casc}(\beta c)$ for $c = \alpha s$.

This computations prove that the semi-Markov renewal kernel of the Markov renewal chain $(J_n, T_n)_{n\geqslant 0}$ is indeed given by (5). Moreover, summing on k in (5) gives

$$\mathbf{P}\left(J_{n+1} = \beta t \middle| J_n = \alpha s\right) = \sum_{\substack{c\in\mathcal{C},\ c=t\ldots \\ c=\ldots[\alpha s]}} \operatorname{casc}\left(\beta c\right) = Q_{\alpha s,\beta t}.$$

The latter provides that $(J_n)_n$ is a Markov process, with transition matrix Q and therefore that Q is stochastic. It gives a proof of Proposition 3.16. □

Proof of Proposition 3.18. Let U be the VLMC defined by $(\mathcal{T}, q)$. As before in the text, for every n, let $C_n = \operatorname{cont}(U_n)$ and Z_n be the alpha-LIS of C_n. Let also J be the internal chain of the semi-Markov process $Z = (Z_n)_n$ – see Definition 3.12 and

Proposition 3.13. Since the context tree is assumed to be stable, any context alpha-LIS is a finite context (Lemma 3.3). Besides, the Markov chain $C = (C_n)_n$ induced on $\mathcal{C}^f$ has been shown to be irreducible (Proposition 3.6). Therefore, two arbitrary finite contexts are joined by an admissible path relative to the Markov chain C. In particular, two arbitrary context alpha-LIS αs and βt are joined by an admissible path relative to the Markov chain C. Taking the alpha-LIS of such a path of contexts provide a path of context alpha-LIS joining αs to βt for the process Z, which means that conditioning by $Z_0 = \alpha s$, there is some $n \geqslant 0$ such that the event $Z_n = \beta t$ occurs with positive probability. Restricting this path to jump times provides an admissible path joining αs to βt relative to the Markov process J. □

Proof of Theorem 3.20 (Invariant probability measures for a stable VLMC).

(*3.* $\Longrightarrow$ *1.*) Since Q is recurrent and irreducible, there exists a unique line $\mathbb{R}v$ of left-fixed vectors for Q. Let $v = (v_{\alpha s})_{\alpha s \in \mathcal{S}}$ be such a vector having non-negative entries (see for example (Seneta, 2006, Theorem 5.4)). Theorem 2.18(ii) coupled with the assumption on the series $\sum_{\alpha s \in \mathcal{S}} v_{\alpha s} \kappa_{\alpha s}$ entails directly the existence and uniqueness of a stationary probability measure.

(*2.* $\Longrightarrow$ *3.*) If there exists a stationary probability measure, then Theorem 2.18(i) and Lemma 2.16 assert that the cascade series converge and that Q admits at least one left-fixed vector v with positive entries such that $\sum_{\alpha s \in \mathcal{S}} v_{\alpha s} \kappa_{\alpha s} < \infty$. Besides, every $\kappa_{\alpha s}$ is greater than 1. Indeed, the cascade of any alpha-LIS is 1 and, in the stable case, any alpha-LIS is a context (Lemma 3.3). Thus, v is summable and Q is positive recurrent (see for instance (Seneta, 2006, Corollary of Theorem 5.5)). Since it is irreducible (Proposition 3.18), it admits a unique direction of left-fixed vectors $\mathbb{R}v$, proving (*3.*) by Theorem 2.18. □

Proof of Theorem 3.25 (finite number of alpha-LIS).

Thanks to Theorem 3.20, (i) and (ii) are equivalent. Moreover, (i) $\Longrightarrow$ (iii) is contained in Theorem 2.18(i). Assume reciprocally that the cascade series converge. Since Q is stochastic, irreducible and finite dimensional, it admits a unique direction of left-fixed vectors, so that Theorem 2.18(ii) allows us to conclude.

Convergence towards π. Denote by $\pi_{\mathcal{C}^f}$ the measure on $\mathcal{C}^f$ induced by π, defined by $\pi_{\mathcal{C}^f}(c) = \pi(c\mathcal{R})$. For any $n \geqslant 0$, denote also by C_n the *cont* of U_n. Thanks to Lemma 2.16(ii), $\pi_{\mathcal{C}^f}$ is a probability measure on $\mathcal{C}^f$. Thus, since the process induced by $(C_n)_n$ on $\mathcal{C}^f$ is an irreducible aperiodic Markov chain (Lemma 3.6) on a denumerable state space that admits $\pi_{\mathcal{C}^f}$ as an invariant probability measure, the distribution of C_n converges to $\pi_{\mathcal{C}^f}$ as soon as C_0 is finite (and $\pi_{\mathcal{C}^f}$ is the unique invariant probability measure on $\mathcal{C}^f$ for the Markov chain induced by $(U_n)_n$ on $\mathcal{C}^f$). In particular, if ν is a distribution of U_0 that does not charge infinite contexts, for any context $c \in \mathcal{C}^f$,

$$\mathbf{P}_\nu\left(U_n \in c\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \pi\left(c\mathcal{R}\right). \tag{6}$$

Now, for any finite word w, use again the notation $w = \beta_1 \cdots \beta_{p_w} \alpha_w s_w$ where the β_k and α_w are letters, p_w is a nonnegative integer and s_w the LIS of w. Take $w \in \mathcal{W}$.

① Assume firstly that w is noninternal and that $n \geqslant p_w$. Then,

$$\mathbf{P}_\nu\left(U_n \in w\mathcal{R}\right) = \operatorname{casc}(w)\,\mathbf{P}_\nu\left(U_{n-p_w} \in \alpha_w s_w \mathcal{R}\right)$$

Since $\alpha_w s_w$ is a context (Lemma 3.3(i)), this entails by (6) that

$$\mathbf{P}_\nu\left(U_n \in w\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \operatorname{casc}(w)\pi\left(\alpha_w s_w \mathcal{R}\right) = \pi\left(w\mathcal{R}\right).$$

② Assume now that w is an internal word. In this case, for any $n \geqslant 0$, by disjoint union, since U_n does not charge any infinite word (because U_0 does not)

$$\mathbf{P}_\nu\left(U_n \in w\mathcal{R}\right) = \sum_{c\in\mathcal{C}^f,\ c=w\cdots} \mathbf{P}_\nu\left(U_n \in c\mathcal{R}\right).$$

Distinguish then the long enough contexts from other ones by defining $\alpha_n(w)$ and $\beta_n(w)$ as the real numbers

$$\alpha_n(w) = \sum_{\substack{c\in\mathcal{C}^f,\ c=w\cdots \\ p_c\leqslant n}} \mathbf{P}_\nu\left(U_n \in c\mathcal{R}\right) \quad \text{and} \quad \beta_n(w) = \sum_{\substack{c\in\mathcal{C}^f,\ c=w\cdots \\ p_c\geqslant n+1}} \mathbf{P}_\nu\left(U_n \in c\mathcal{R}\right).$$

Begin with $\alpha_n(w)$ that can be written,

$$\alpha_n(w) = \sum_{\alpha s\in\mathcal{S}} \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s] \\ p_c\leqslant n}} \mathbf{P}_\nu\left(U_n \in c\mathcal{R}\right) = \sum_{\alpha s\in\mathcal{S}} \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s] \\ p_c\leqslant n}} \operatorname{casc}(c)\,\mathbf{P}_\nu\left(U_{n-p_c} \in \alpha s\mathcal{R}\right). \tag{7}$$

For a given $\alpha s \in \mathcal{S}$, by hypothesis, the cascade series

$$\sum_{\substack{c\in\mathcal{C}^f \\ c=\cdots[\alpha s]}} \operatorname{casc}(c)$$

converges. Thus, the convergence in the last sum of (7) is dominated so that, using (6) again,

$$\sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s] \\ p_c\leqslant n}} \operatorname{casc}(c)\,\mathbf{P}_\nu\left(U_{n-p_c} \in \alpha s\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s]}} \operatorname{casc}(c)\pi\left(\alpha s\mathcal{R}\right).$$

Since $\mathcal{S}$ is finite, one gets finally

$$\alpha_n(w) = \sum_{\alpha s\in\mathcal{S}} \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s] \\ p_c\leqslant n}} \operatorname{casc}(c)\,\mathbf{P}_\nu\left(U_{n-p_c} \in \alpha s\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \sum_{\alpha s\in\mathcal{S}} \sum_{\substack{c\in\mathcal{C}^f \\ c=w\cdots=\cdots[\alpha s]}} \operatorname{casc}(c)\pi\left(\alpha s\mathcal{R}\right) = \pi\left(w\mathcal{R}\right).$$

③ Extend now the definition of the α_n and β_n to any finite word, by denoting

$$\alpha_n(w) = \mathbf{P}_\nu\left(U_n \in w\mathcal{R}\right) \text{ and } \beta_n(w) = 0$$

whenever w is noninternal. With this notation, if w is *any* finite word

$$\mathbf{P}_\nu\left(U_n \in w\mathcal{R}\right) = \alpha_n(w) + \beta_n(w) \tag{8}$$

and it is shown in ① and ② that

$$\alpha_n(w) \underset{n\to\infty}{\longrightarrow} = \pi\left(w\mathcal{R}\right) \text{ and } \beta_n(w) \geqslant 0. \tag{9}$$

Take finally any finite word w and denote its length by N. By disjoint union, if $n \geqslant 1$,

$$1 = \sum_{v\in\mathcal{W},\ |v|=N} \mathbf{P}_\nu\left(U_n \in v\mathcal{R}\right) = \sum_{v\in\mathcal{W},\ |v|=N} \alpha_n(v) + \sum_{v\in\mathcal{W},\ |v|=N} \beta_n(v). \tag{10}$$

Using (9), this sum being finite, one gets

$$\sum_{v\in\mathcal{W},\ |v|=N} \alpha_n(v) \underset{n\to\infty}{\longrightarrow} \sum_{v\in\mathcal{W},\ |v|=N} \pi\left(v\mathcal{R}\right) = 1, \tag{11}$$

the last equality resulting from disjoint union. Putting (10) and (11) together shows that

$$\lim_{n\to\infty} \sum_{v\in\mathcal{W},\ |v|=N} \beta_n(v) = 0.$$

In particular, if w is *any* finite word,

$$\lim_{n\to\infty} \beta_n(w) = 0. \tag{12}$$

Thus, (8), (9) and (12) show the result. □

Proof of Theorem 3.29 (A SMC on a finite state space is a stable VLMC).

Let $(Z_n)_{n\geqslant 0}$ be a semi-Markov chain with true jumps on a state space of cardinality b. Without lack of generality, for technical convenience, take $\mathcal{A} = \mathbb{Z}/b\mathbb{Z}$. Denote by $(J_N)_{N\geqslant 0}$ the internal chain of $(Z_n)_n$, namely the process of the successive *different* states of $(Z_n)_{n\geqslant 0}$ and by $(T_N)_{N\geqslant 0}$ the process of the successive sojourn times in the visited states, with the convention $T_0 = 0$. Assuming that $(Z_n)_{n\geqslant 0}$ is semi-Markov amounts to supposing that almost surely, for every $N \geqslant 0$, $\alpha \in \mathcal{A}$, $k \geqslant 1$,

$$\mathbf{P}\left(J_{N+1} = \alpha, T_{N+1} = k \middle| J_0, \cdots, J_N, T_1, \cdots, T_N\right) = \mathbf{P}\left(J_{N+1} = \alpha, T_{N+1} = k \middle| J_N\right).$$

Denote by $p = (p_{\alpha,\beta}(k))_{\alpha,\beta\in\mathcal{A},k\geqslant 1}$ the semi-Markov kernel of the process:

$$p_{\alpha,\beta}(k) = \mathbf{P}\left(J_{N+1} = \beta, T_{N+1} = k \middle| J_N = \alpha\right).$$

For any $n \geqslant 0$, define the right-infinite random sequence

$$V_n = Z_n Z_{n-1} \cdots Z_1 Z_0 \, (1 + Z_0) \, Y_{-2},$$

where Y_{-2} is any non atomic distribution on $\mathcal{R}$. Remember that the process $(Z_n)_{n \geqslant 0}$ takes its values in the set $\mathcal{A} = \mathbb{Z}/b\mathbb{Z}$. The semi-Markov property of $(Z_n)_{n \geqslant 0}$ guarantees that the $\mathcal{R}$-valued process $(V_n)_{n \geqslant 0}$ is Markovian.

Define now $(U_n)_{n \geqslant 0}$ as the VLMC on the b-comb with alphabet $\mathcal{A}$, with the following parameters. As an initial state, take $U_0 = Z_0 \, (1 + Z_0) \, Y_{-2}$. As a transition probability associated with the context $\beta^\ell \gamma$ where $\beta, \gamma \in \mathcal{A}$, $\beta \neq \gamma$ and $\ell \geqslant 1$, take the probability measure $q_{\beta^\ell \gamma}$ on $\mathcal{A}$:

$$\forall \alpha \in \mathcal{A}, q_{\beta^\ell \gamma}(\alpha) = \begin{cases} \dfrac{p_{\beta,\alpha}(\ell)}{\displaystyle\sum_{k \geqslant \ell} \left(\sum_{\delta \in \mathcal{A}} p_{\beta,\delta}(k) \right)} \text{ if } \alpha \neq \beta \\ \\ 1 - \dfrac{\displaystyle\sum_{\delta \in \mathcal{A}} p_{\beta,\delta}(\ell)}{\displaystyle\sum_{k \geqslant \ell} \left(\sum_{\delta \in \mathcal{A}} p_{\beta,\delta}(k) \right)} = \dfrac{\displaystyle\sum_{k \geqslant \ell+1} \left(\sum_{\delta \in \mathcal{A}} p_{\beta,\delta}(k) \right)}{\displaystyle\sum_{k \geqslant \ell} \left(\sum_{\delta \in \mathcal{A}} p_{\beta,\delta}(k) \right)} \text{ if } \alpha = \beta. \end{cases} \tag{13}$$

Note that $q_{\beta^\ell \gamma}$ does not depend on γ and that the convergence of the series is guaranteed by the properties of the semi-Markov kernel p.

We show that both $\mathcal{R}$-valued Markov processes $(U_n)_{n \geqslant 0}$ and $(V_n)_{n \geqslant 0}$ have the same distribution stating that, for any $\alpha \in \mathcal{A}$ and $n \geqslant 0$, almost surely,

$$\mathbf{P}\left(V_{n+1} = \alpha V_n \middle| V_n\right) = q_{\mathrm{cont}(V_n)}(\alpha).$$

Since V_0 and U_0 have the same distribution, this will entail the result.

Let $r \in \mathcal{R}$, not of the form β^∞, $\beta \in \mathcal{A}$; write $r = \beta^\ell s$ where $\ell \geqslant 1$, $\beta \in \mathcal{A}$ and $s \in \mathcal{R}$ starts with a letter different from β. Denote, by S_n the partial sum of T_k, as in Definition 3.12. For any $\alpha \in \mathcal{A}$, $\alpha \neq \beta$ and for any $n \geqslant 0$, as soon as N is such that $S_N \leqslant n \leqslant S_{N+1} - 1$,

$$\mathbf{P}\left(V_{n+1} = \alpha V_n \middle| V_n = r\right) = \mathbf{P}\left(J_{N+1} = \alpha, T_{N+1} = \ell \middle| J_N = \beta, T_{N+1} \geqslant \ell, V_{n-\ell} = s\right).$$

Because of the semi-Markov property, this entails that

$$\mathbf{P}\left(V_{n+1} = \alpha V_n \middle| V_n = r\right) = \mathbf{P}\left(J_{N+1} = \alpha, T_{N+1} = \ell \middle| J_N = \beta, T_{N+1} \geqslant \ell\right).$$

Using the semi-Markov kernel, this leads to

$$\mathbf{P}\left(V_{n+1} = \alpha V_n \middle| V_n = r\right) = \frac{\mathbf{P}\left(J_{N+1} = \alpha, T_{N+1} = \ell \middle| J_N = \beta\right)}{\mathbf{P}\left(T_{N+1} \geqslant \ell \middle| J_N = \beta\right)} = \frac{p_{\beta,\alpha}(\ell)}{\displaystyle\sum_{\delta \in \mathcal{A}} \sum_{k \geqslant \ell} p_{\beta,\delta}(k)}.$$

This shows that Formulae (13) are the suitable ones to ensure that both Markov processes $(U_n)_{n\geqslant 0}$ and $(V_n)_{n\geqslant 0}$ have the same distribution. □

Proof of Theorem 3.30 (Limit distribution of a semi-Markov chain).

Let b be the cardinal of the state space $\mathcal{E}$. Let U be the VLMC defined on the b-comb (Definition 3.28) by the transition probabilities of Formula (13). This context tree (the b-comb) is defined on the alphabet $\mathcal{A} = \mathcal{E}$; it admits a finite number of context alpha-LIS, namely the words $\alpha\beta$ where α and β are two distincts elements of $\mathcal{E}$; furthermore, in this tree, the set of contexts having $\alpha\beta$ as an alpha-LIS is $\left\{\alpha^\ell\beta :\ \ell \geqslant 1\right\}$. The processes $(Z_n)_n$ and U are related by Theorem 3.29. Moreover, Theorem 3.25 applies to U. Indeed, thanks to Formulas (13), the assumption on the positivity of the semi-Markov kernel p implies that the associated VLMC U is non-null.

By Theorem 3.25, the convergence of the cascade series implies the convergence $\mathbf{P}\left(U_n \in w\mathcal{R}\right) \to \pi\left(w\mathcal{R}\right)$ when n tends to infinity, for any finite word w. As a consequence, as soon as the cascade series converge, $\mathbf{P}\left(Z_n = \beta | Z_0 = \alpha\right)$ converges to $\pi\left(\beta\mathcal{R}\right)$ when n tends to infinity, and this limit does not depend on α. In other words, $(Z_n)_{n\geqslant 0}$ admits $\left(\pi\left(\beta\mathcal{R}\right)\right)_{\beta\in\mathcal{E}}$ as a limit distribution.

Conversely, assume that $(Z_n)_n$ admits a limit distribution μ. In terms of the VLMC $(U_n)_n$, this implies that $\mathbf{P}\left(U_n \in \alpha\mathcal{R}\right)$ tends to $\mu(\alpha)$ when n tends to infinity, for every $\alpha \in \mathcal{E}$. Let $\alpha, \beta \in \mathcal{E}$ be two distinct letters, so that $\alpha\beta \in \mathcal{S}$. For every $n \geqslant 1$,

$$\mathbf{P}\left(U_n \in \alpha\beta\mathcal{R}\right) = \mathbf{P}\left(Z_n = \alpha, Z_{n-1} = \beta\right) = p_{\beta,\alpha}(1)\,\mathbf{P}\left(Z_{n-1} = \beta\right)$$

so that

$$\mathbf{P}\left(U_n \in \alpha\beta\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} p_{\beta,\alpha}(1)\mu(\beta).$$

This has two consequences.

Firstly, $\mu(\alpha) \neq 0$ for every $\alpha \in \mathcal{E}$. Indeed, since μ is a probability measure, all these numbers cannot simultaneously vanish; let thus $\beta \in \mathcal{E}$ such that $\mu(\beta) \neq 0$. Let also $\alpha \in \mathcal{E} \setminus \{\beta\}$. For any $n \geqslant 1$, as before,

$$\mathbf{P}\left(Z_n Z_{n-1} = \alpha\beta\right) = p_{\beta,\alpha}(1)\,\mathbf{P}\left(Z_{n-1} = \beta\right).$$

Assume that $\mu(\alpha) = 0$. Since $\mathbf{P}\left(Z_n Z_{n-1} = \alpha\beta\right) \leqslant \mathbf{P}\left(Z_n = \alpha\right)$, the left hand side of this equality tends to 0 when n tends to infinity, which is impossible because the limit right hand side, namely $p_{\beta,\alpha}(1)\mu(\beta)$, is non-null. Thus, $\mu(\alpha) \neq 0$.

Secondly, for every $\ell \geqslant 1$,

$$\mathbf{P}\left(U_n \in \alpha^\ell\beta\mathcal{R}\right) = \operatorname{casc}\left(\alpha^\ell\beta\right)\mathbf{P}\left(U_{n-l+1} \in \alpha\beta\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \operatorname{casc}\left(\alpha^\ell\beta\right) p_{\beta,\alpha}(1)\mu(\beta).$$

Thus, if L is a positive integer,

$$\sum_{\ell=1}^{L} \mathbf{P}\left(U_n \in \alpha^\ell\beta\mathcal{R}\right) \underset{n\to\infty}{\longrightarrow} \left[\sum_{\ell=1}^{L} \operatorname{casc}\left(\alpha^\ell\beta\right)\right] p_{\beta,\alpha}(1)\mu(\beta). \tag{14}$$

For any n, the left hand side of Formula (14) if less than 1 because it is bounded above by the whole sum $\sum_{c\in\mathcal{C}^f}\mathbf{P}\left(U_n\in c\mathcal{R}\right)+\sum_{c\in\mathcal{C}^i}\mathbf{P}\left(U_n=c\right)$ which equals 1 by disjoint union. Thus, the right hand side is also bounded. In particular, since $p_{\beta,\alpha}(1)$ and $\mu(\beta)$ are non-null, the series of positive numbers $\sum_\ell \operatorname{casc}\left(\alpha^\ell\beta\right)$ converge. Besides, the cascade series of the context alpha-LIS $\alpha\beta$ is precisely $\kappa_{\alpha\beta}=\sum_{\ell\geqslant 1}\operatorname{casc}\left(\alpha^\ell\beta\right)$. This allows us to conclude that the existence of a limit distribution for the process $\left(Z_n\right)_n$ implies the convergence of all the cascade series.

It suffices now to show that condition (ii) is equivalent to the convergence of every cascade series

$$\sum_{\ell\geqslant 1}\operatorname{casc}\left(\alpha^\ell\beta\right).$$

Let α and β be two distinct elements of $\mathcal{A}$. For any $\ell\geqslant 1$, since

$$\operatorname{casc}\left(\alpha^\ell\beta\right)=\prod_{k=1}^{\ell-1}q_{\alpha^k\beta}(\alpha),$$

an immediate reading of Formula (13) shows that

$$\operatorname{casc}\left(\alpha^\ell\beta\right)=\frac{\displaystyle\sum_{k\geqslant\ell}\left(\sum_{\gamma\in\mathcal{A}}p_{\alpha,\gamma}(k)\right)}{\displaystyle\sum_{k\geqslant 1}\left(\sum_{\gamma\in\mathcal{A}}p_{\alpha,\gamma}(k)\right)},$$

so that the sum of these cascades writes

$$\sum_{\ell\geqslant 1}\operatorname{casc}\left(\alpha^\ell\beta\right)=\frac{\displaystyle\sum_{k\geqslant 1}k\left(\sum_{\gamma\in\mathcal{A}}p_{\alpha,\gamma}(k)\right)}{\displaystyle\sum_{k\geqslant 1}\left(\sum_{\gamma\in\mathcal{A}}p_{\alpha,\gamma}(k)\right)}\in[0,+\infty],$$

leading to the result. Note that the cascade series $\sum_{\ell\geqslant 1}\operatorname{casc}\left(\alpha^\ell\beta\right)$ does not depend on β, because all the probability measures one has to associate with the leaves of a given context of the comb are all the same ones, as can be seen on Formula (13). □

2. Appendix: an example of invariant σ-finite measure that charges irrational infinite contexts

A soon as a non-null VLMC admits an invariant probability measure, all infinite words are negligible – see Lemma 2.16(ii). Besides, the same argument as in the proof of that

lemma shows that an invariant σ-finite measure always vanishes on *rational* right-infinite words, *i.e.* on eventually periodic words. This appendix provides an example of non-null VLMC that admits an invariant σ-finite measure which gets positive values on infinitely many (necessarily irrational) contexts[1].

In this appendix, a "σ-finite measure" denotes a positive non-zero measure on $\mathcal{R}$ which is finite on all cylinders $c\mathcal{R}$ based on finite contexts c, and also, necessarily, on infinite contexts. In particular, since the contexts induce a partition of $\mathcal{R}$, such a measure is truly σ-finite. As usual in the field of Markov chains, when U is a VLMC, the definition of an U-invariant probability measure can be extended to σ-finite measures using the transition probability kernel P_U, defined by

$$P_U(r, B) = \sum_{\alpha \in \mathcal{A}} q_{\mathrm{cont}(r)}(\alpha) \mathbb{1}_{\{\alpha r \in B\}}$$

on Borel sets B and right-infinite words r: this kernel acts on σ-finite measures π through the formula

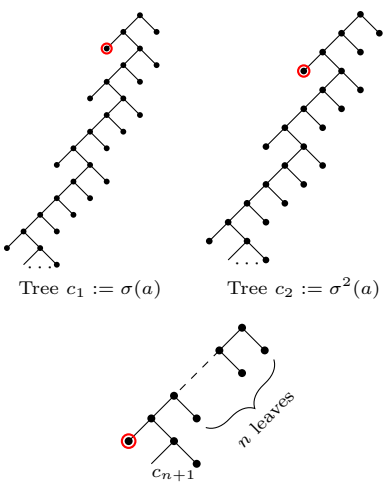
$$\pi P_U(B) := \int_{\mathcal{R}} P_U(x, B) d\pi(x)$$

(this is an action on the right), and a σ-finite measure π is said U-invariant whenever $\pi P_U = \pi$.

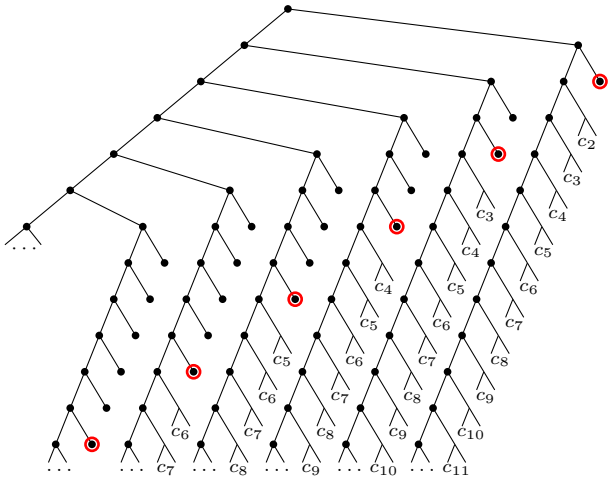
In what follows, we describe the announced example in the form of a sequence of hints and assertions that can be easily (but sometimes laboriously) verified.

Consider the irrational right-infinite word $a = 1010^2 10^3 10^4 \dots$. Define as follows *the stabilised arithmetic tree*, denoted by $\mathcal{T}_a$: it is the *stable* context tree on the alphabet $\mathcal{A} = \{0, 1\}$ spanned by a, *i.e.* the smallest context tree that contains all the shifted words $\sigma^n(a)$, $n \geqslant 0$ (see Definition 2.6). On the left side of the following picture, one can find drawings of the successive context trees t_n spanned by the shifted infinite word $\sigma^n(a)$, $n \geqslant 1$. They are used to give a representation of $\mathcal{T}_a$ on the right side of the picture.

[1] In particular, one cannot get rid of the finiteness assumption of an invariant measure to prove Lemma 2.16(ii).



Tree $c_n = \sigma^n(a),\ n \geqslant 1$



Stable context tree $\mathcal{T}_a$, generated by $a = 1010^2 10^3 \ldots$

Computing the contexts leads to show that $\mathcal{T}_a$ admits two one-parameter families of context alpha-LIS, namely $0^q 10^q 1$, $q \geqslant 0$ and $10^N 10^{N+2}$, $N \geqslant 1$. They are coloured red in the pictures. Thus the related matrix Q is infinite – see (7) for a definition of Q. Moreover, the infinite contexts of $\mathcal{T}_a$ are the following ones: on one side, the 0^∞ and the $0^n 10^\infty$, $n \geqslant 0$ which are rational; on the other side, the $\sigma^n(a)$ which are irrational.

Given a non-null probabilising of $\mathcal{T}_a$, assume that an invariant σ-finite measure μ exists. Reasoning like in the proof of Theorem 2.18, for every $\alpha s \in \mathcal{S}$, decompose the number $\mu(\alpha s \mathcal{R})$ through the partition of $s\mathcal{R}$ induced by cylinders based on finite contexts that have s as a prefix and by such infinite contexts as well. These writings show that Q's entries are necessarily finite (sums of summable families) as well as the sums $\sum_{c \in \mathcal{C}^i,\ c=s\ldots} q_c(\alpha)\mu(c)$. Note further that Q is row-stochastic because $\mathcal{T}_a$ is a stable tree (see Proposition 3.16). Since all infinite contexts are shifted from a and since μ is invariant, all the numbers $\mu(c)$, $c \in \mathcal{C}^i$ can be written as $\mu(c) = \frac{\mu(a)}{p_c}$ where p_c is a finite product of $q_{\sigma^k(a)}(\beta)$, $k \geqslant 0$, $\beta \in \mathcal{A}$. Consequently, all the sums

$$\ell_{\alpha s} := \sum_{\substack{c \in \mathcal{C}^i \\ c=s\ldots}} \frac{q_c(\alpha)}{p_c}$$

are also finite – note that the sums $\ell_{0^q 10^q 1}$, $q \geqslant 1$ are reduced to a single term, all other $\ell_{\alpha s}$'s being true infinite sums.

Finally, like in the proof of Theorem 2.18, the above decompositions lead to the following statement: a σ-finite measure μ is invariant if and only if it satisfies the (infinite) matricial equation

$$\mu(a)\ell + \mu_{\mathcal{S}} Q = \mu_{\mathcal{S}} \tag{15}$$

where ℓ and $\mu_{\mathcal{S}}$ denote the infinite row-vectors $\ell = (\ell_{\alpha s})_{\alpha s \in \mathcal{S}}$ and $\mu_{\mathcal{S}} = (\mu(\alpha s \mathcal{R}))_{\alpha s \in \mathcal{S}}$. Notice that after a straightforward continuation of the function f to σ-finite measures,

$\mu_{\mathcal{S}} = f(\mu)$ (see (2.4)). Using the vocabulary of Kitchens (1998), one finally gets the following result.

Proposition 2.1. *Let U be some non-null VLMC defined from the context tree $\mathcal{T}_a$ and let Q be its Q-matrix (see (7)).*

(i) If Q is positive recurrent, then U admits a unique half-line of invariant σ-finite measures. All of them are finite ones.

(ii) If Q is null recurrent, then U admits a unique half-line of invariant σ-finite measures. None of them are finite ones, but they turn all infinite words negligeable.

(iii) If Q is transient, then U does not admit any invariant probability measure.

We build an example of VLMC on $\mathcal{T}_a$ that admits an invariant σ-finite measure that charges (all) irrational infinite contexts. Proposition 2.1 shows that the corresponding Q-matrix is necessarily transient.

To exhibit such an example, one first have to compute the "form" of Q (check which entries vanish, see below) and to make explicit the way how ℓ's and Q's entries are expressed in terms of the q_c. This being done, one sees that any row-stochastic matrix A having the form of Q (same positive entries, same zero ones) is the Q-matrix of a probabilised context tree $\mathcal{T}_a$ (choice of the q_c, $c \in \mathcal{C}^f$). Furthermore, if X is any positive row-vector that satisfies $XA < X$ (strict inequality for every coordinate), the vector $X - XA$ can be chosen as the ℓ-vector of such a probabilised $\mathcal{T}_a$ (choice of the q_c, $c \in \mathcal{C}^i$).

Therefore, thanks to Equation (15), an example of invariant σ-finite measure that charges (all) irrational infinite contexts is given by any row-stochastic matrix A having the required form, together with a positive row-vector X that satisfies $XA < X$. Note that such an A is necessarily transient and that a corresponding X has necessarily non summable coordinates (see Kitchens (1998)). Below, we give such a matrix A.

Order the context alpha-LIS by increasing length, placing $10^{q-1}10^{q+1}$ before 0^q10^q1 (both alpha-LIS have the same length). For this order, the form of Q is written hereunder, a $*$ denoting a positive entry. As a heuristic hint, observe firstly that the (transient) matrix V given hereunder and the positive vector $X = (1, 3, 2, 5, 4, 7, 6, 9, 8 \dots)$ satisfy

$XV < X$.

$$Q = \begin{pmatrix}
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & 0 & * & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & 0 & * & 0 & * & 0 & 0 & 0 & 0 & 0 & \dots \\
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & 0 & * & 0 & * & 0 & * & 0 & 0 & 0 & \dots \\
* & * & * & * & * & * & * & * & * & * & * & \dots \\
* & * & 0 & * & 0 & * & 0 & * & 0 & * & 0 & \dots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{pmatrix}
\quad V = \begin{pmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{pmatrix}$$

The (transient) matrix A we give is a deformation of V that has the form of Q. It satisfies $XA < X$ for the row-vector X given above. Let $r, s \in]0,1[$ and let A be the matrix

$$A = \begin{pmatrix}
\frac{r^2}{0!} & \frac{r^3}{1!} & R_1(r) & \frac{r^5}{3!} & \frac{r^6}{4!} & \frac{r^7}{5!} & \frac{r^8}{6!} & \frac{r^9}{7!} & \frac{r^{10}}{8!} & \frac{r^{11}}{9!} & \frac{r^{12}}{10!} & \cdots \\
\frac{r^3}{0!} & \frac{r^4}{1!} & \frac{r^5}{2!} & \frac{r^6}{3!} & R_2(r) & \frac{r^8}{5!} & \frac{r^9}{6!} & \frac{r^{10}}{7!} & \frac{r^{11}}{8!} & \frac{r^{12}}{9!} & \frac{r^{13}}{10!} & \cdots \\
s & 1-s & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
\frac{r^5}{0!} & \frac{r^6}{1!} & \frac{r^7}{2!} & \frac{r^8}{3!} & \frac{r^9}{4!} & \frac{r^{10}}{5!} & R_4(r) & \frac{r^{12}}{7!} & \frac{r^{13}}{8!} & \frac{r^{14}}{9!} & \frac{r^{15}}{10!} & \cdots \\
\frac{s^2}{2!} & \frac{s^2}{2!} & 0 & 1-\frac{s^2}{1!} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\
\frac{r^7}{0!} & \frac{r^8}{1!} & \frac{r^9}{2!} & \frac{r^{10}}{3!} & \frac{r^{11}}{4!} & \frac{r^{12}}{5!} & \frac{r^{13}}{6!} & \frac{r^{14}}{7!} & R_6(r) & \frac{r^{16}}{9!} & \frac{r^{17}}{10!} & \cdots \\
\frac{s^3}{3!} & \frac{s^3}{3!} & 0 & \frac{s^3}{3!} & 0 & 1-\frac{s^3}{2!} & 0 & 0 & 0 & 0 & 0 & \cdots \\
\frac{r^9}{0!} & \frac{r^{10}}{1!} & \frac{r^{11}}{2!} & \frac{r^{12}}{3!} & \frac{r^{13}}{4!} & \frac{r^{14}}{5!} & \frac{r^{15}}{6!} & \frac{r^{16}}{7!} & \frac{r^{17}}{8!} & \frac{r^{18}}{9!} & R_8(r) & \cdots \\
\frac{s^4}{4!} & \frac{s^4}{4!} & 0 & \frac{s^4}{4!} & 0 & \frac{s^4}{4!} & 0 & 1-\frac{s^4}{3!} & 0 & 0 & 0 & \cdots \\
\frac{r^{11}}{0!} & \frac{r^{12}}{1!} & \frac{r^{13}}{2!} & \frac{r^{14}}{3!} & \frac{r^{15}}{4!} & \frac{r^{16}}{5!} & \frac{r^{17}}{6!} & \frac{r^{18}}{7!} & \frac{r^{19}}{8!} & \frac{r^{20}}{9!} & \frac{r^{21}}{10!} & \cdots \\
\frac{s^5}{5!} & \frac{s^5}{5!} & 0 & \frac{s^5}{5!} & 0 & \frac{s^5}{5!} & 0 & \frac{s^5}{5!} & 0 & 1-\frac{s^5}{5!} & 0 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{pmatrix}$$

where $R_1(r) = 1 + \frac{r^4}{2} - r^2 e^r$ and $R_n(r) = 1 + \frac{r^{2n+3}}{(n+2)!} - r^{n+1} e^r$ when $n \geqslant 2$ so that A is row-stochastic. It turns out that A satisfies $XA < X$ as soon as r and s are small enough. More precisely, $0 < r < 1/3$ and $0 < s < 1/10$ is a sufficient condition. All this can be checked by patient but simple calculations.

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